

BIO 1114 – PLANT DIVERSITY AND ECOLOGY

Fall 2026 Syllabus

Lecture: Mon. and Wed. 4:30pm – 5:50pm, Building 180 (Baker Science) Room 101

Lab: Building 33 (Fisher Science) Room 351 - All Lab Sections

Reed Kenny (Lecture Instructor, Lab Coordinator)

Office: 53-318

Office Hours: M 10-11, T 3-4, W 3-4, and by appointment

Email: rekenny@calpoly.edu

Lab Sections

Section	Time	Room	Instructor	Email address	Office	Office Hours
02	W 9-11:50am	33-351	Reed Kenny	rekenny@calpoly.edu	53-318	M 10-11, T 3-4, W 3-4
03	W 12-2:50pm	33-351	Ben Sherman			
04	Th 9-11:50am	33-351	Mike Butler			
05	Th 12-2:50pm	33-351	Mike Butler			

Course Schedule:

Week	Lecture Topic	Lab Topic	Lab Exercise
1	8/24: Introduction; Field Trip Information	Propagate Ferns; Leaf Collection Assignment; Vegetative Features, Intro to campus trees	Ex. 1, 6, 7. Assign Vegetative Features
	8/26: Leaf Morphology		
2	8/31: California Ecosystems and Floristics, Plant Names	Field Trip: Horse Canyon	Field Notebook Guide; Field Trip Guide
	9/2: Plant Parts Continued		
3	9/7: (Labor Day)	Using ID Keys; Visit Hoover Herbarium, mini field trip to Poly Canyon. Vegetative Features Due	Ex. 8, 14. Assign Using ID Keys, Plant collection, Herbarium questions
	9/9: Using Identification Keys; Plant Collections and Herbaria		
4	9/14: Plants and plant communities on the dunes	Field Trip: Los Osos Dunes FIELD QUIZ #1	Field Trip Guide
	LECTURE EXAM #1 (September 16th)		
5	9/21: Plant Reproduction Part 1	Flowers and Fruit, Herbarium Questions Due	Ex. 10 and 11 Assign Grocery store botany
	9/23: Plant Reproduction Part 2		
6	9/28: Oak woodlands	Field trip: Bishop Peak	Field Trip Guide
	9/30: Ferns, Club Mosses, and Horsetails		

7	10/5: Fern ecology and diversity	Ferns, Club Mosses and Horsetails	Ex. 15
	10/7: Coastal scrub and mediterranean diversity		
8	10/16: Review	Field Trip: Elfin Forest	Field Trip Guide
	10/14: LECTURE EXAM #2 (October 14)		
9	10/19: Gymnosperm reproduction	LAB MIDTERM EXAM (October 21/22) Field Notebooks Due	Ex. 16, Campus Conifers Tour
	10/21: Gymnosperm diversity and ecology		
10	10/26: Fig tree biology	Fig Tree Biology	Ex. 12
	10/28: Plant interactions, competition and mutualism		
11	11/2: Saltwater marshes and the plants that occur in them	Field trip: Sweet Springs Reserve, Los Osos. FIELD QUIZ #2	Field Trip Guide
	11/4: Deserts and drought adaptations		
12	11/9: Chaparral	No Lab – Veterans Day Nov. 11.	
	11/11: (Veterans Day)		
13	11/16: Tropical Forest Biology	Tour Cal Poly Plant Conservatory	Ex. 17
	11/18: Economically valuable plants/Thanksgiving Ethnobotany		
14	11/23: Fall Break		
	11/25: Fall Break		
15	11/30: Grasslands and Wildflowers	Review Field Trip Plants, Mount Specimens	
	12/2: Seed ecology and dispersal		
16	12/7: Habitat Restoration	LAB FINAL EXAM (December 9/10th) Field Notebooks Due	
	12/9: Humans and Plants		

LECTURE EXAM #1: WEDNESDAY SEPTEMBER 16th 4:30-5:50pm

LECTURE EXAM #2: WEDNESDAY OCTOBER 14th 4:30-5:50pm

LECTURE FINAL EXAM: FRIDAY DECEMBER 18th 1:00-3:30pm

Learning Objectives:

Students who complete Biology 114 with a passing grade are expected to have a better understanding of the following:

1. The morphology of plant roots, shoots, flowers, fruits, and leaves
2. The effects of California climate and topography on vegetation
3. The dominant species composition for the most common California plant communities (chaparral, coastal scrub, riparian, oak woodlands, grasslands, etc.)
4. The skills necessary for plant recognition and identification, including the use and creation of dichotomous plant identification keys
5. The names of many of the dominant plants in the local plant communities
6. The basic skills of keeping a field notebook
7. The basic skills of plant collection and herbarium use
8. Plant evolution and diversity
9. The natural world and the importance of plants in providing habitat and nourishment for most animals

Required Texts:

California Plants, by Matt Ritter, Pacific Street Publishing, 2018

Botany for Gardeners, 4th Edition, by Brian Capone, Timber Press

Plant Biology: Lab Manual and Field Trip Guides, 2nd Edition, by Matt Ritter, Pacific Street Publishing, 2017

A Californian's Guide to the Trees Among Us, Expanded and Updated, by Matt Ritter, Heyday, 2022

For your field notebook, purchase a lined black and white bound [composition notebook](#)

Disability Resource Center:

It is University policy to provide reasonable accommodations to students who have disabilities that may affect their ability to participate in course activities or to meet course requirements. If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact the Disability Resource Center at (805) 756-1395 or via email at drc@calpoly.edu, as early as possible in the term.

Academic Integrity at Cal Poly

You are expected to uphold the academic integrity of Cal Poly State University, San Luis Obispo while you are a student. This means that you do your own work, do not plagiarize anyone else's work, or use information from other outside sources, including artificial intelligence (AI) generated materials without properly citing the source(s). Do not inform your fellow classmates about exam or quiz questions, or obtain exams, quizzes, assignments, or lab reports from previous terms. Do not glance at someone else's exam or quiz, copy their answers, or share your answers with anyone. If you are caught cheating, you will receive zero points for that assignment, quiz, or exam. If the infraction is deemed egregious, you will be reported to the Office of Student Rights & Responsibilities. You may review the policies regarding academic integrity here: <https://osrr.calpoly.edu/academic-integrity>

Field Trips Rules

- Drive slowly and carefully to the field trip sites. This is the most dangerous part of the class!
- Arrive to the field trip site on time (field trip start times are announced during the previous lecture, posted on Canvas, and via email). Text your instructor if you will be late. Don't speed! Get there safely first and foremost.
- Wear appropriate clothing: sturdy shoes, long pants, a hat, warm clothes, and layers
- Stay curious and ask lots of questions
- Stay close to your instructor, pay close attention
- Take good notes in your field notebook
- Stay positive and help your fellow classmates

Course Grading:

Lecture (300 points total)

Lecture Exam #1: 100 points, held on **Wednesday, September 16th 4:30-5:50 PM**

Lecture Exam #2: 100 points, held on **Wednesday, October 21st 4:30-5:50 PM**

Lecture Final: 100 points, held on **Friday, December 18th 1:00 PM**

Material discussed in lecture and material from assigned readings will be on the exams. All exams are cumulative, but stress the most recent material covered.

Laboratory (400 points total)

Lab Midterm Exam #1: 50 points, held on **Wed/Thurs September 23/24th**

Lab Midterm Exam #2: 50 points, held on **October 28/29th**

Lab Final Exam: 100 points, held on **Wed/Thurs December 9/10th**

Attendance at field trips: 35 points (Roll will be taken on all field trips)

Field Notebook: 50 points

Quizzes and assignments: 115 points, broken down as follows:

(2) Field Quizzes (10 points each)	20 points
Ex. 6 - Leaf Collection Assignment:	10 points
Ex. 7 - Vegetative Features:	20 points
Ex. 8 - Tree Identification:	10 points
Ex. 10 - Grocery Store Botany Assignment:	10 points
Ex. 14 - Herbarium Plant Collections:	20 points
Ex. 14 - Herbarium Study Questions:	10 points
Ex. 17 - Plant Conservatory Assignment:	15 points

Final Grade: Lecture will account for 50% of your grade. Lab will count for 50% of your grade. Grade scales follow a (+) and (-) calculation as follows:

93.0 - 100%	A
90.0 - 92.9%	A-
87.0 - 89.9	B+
83.0 - 86.9%	B
80.0 - 82.99%	B-
77.0 - 79.9%	C+
73.0 - 76.9%	C
70.0 - 72.9%	C-
67.0 - 69.9%	D+
63.0 - 66.9%	D
60.0 - 62.9%	D-
< 60.0%	F

***To calculate your current overall grade during the course, average the Lecture % and Lab % posted on the Canvas Gradebook.**

NOTE: The Canvas Total Score is NOT correct.